

The Evolution of 3-Point Shot & the Rise of the Small-Ball Era

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Topic

How has the marginal effect of 3-point attempt rate on team winning outcomes evolved overtime from the 2005 to 2024 NBA seasons? We want to capture the potential nonlinear changes in the marginal effect of three-point attempt rate on team performance, reflecting the rise and maturation of the small-ball era.

Regression Model

$$WinRate_{it} = \alpha_i + \gamma_t + \beta_1 3PA\%_{it} + \beta_2 (3PA\%_{it} \times YearIndex_t) + \beta_3 (3PA\%_{it} \times YearIndex_t^2) + X_{it} + \epsilon \tag{1}$$

We form a quadratic interaction OLS regression model using panel data with team and season observations. The dependent variable is team winning percentage (WinRate). The main independent variable is the three-point attempt rate (3PA%), calculated as the share of three-point attempts in total field-goal attempts. Team and season fixed effects are included in control of time-invariant and team-wide characteristics. Control variables like offensive rating, defensive rating, and game pace are incorporated to isolate the effect of 3PA% on team performance. Definitions of all variables used in the model are provided in Table 1.

Table 1. Description of Variables

<i>Symbol</i>	<i>Description</i>
i	Team index
t	Time index (season)
$WinRate$	Winning percentage
$3PA\%$	3-point attempt rate = 3PA / FGA
α_i	Team fixed effect—controls for time-invariant characteristics
γ_t	Season fixed effect—controls for effects of all teams within a year (season)
X_{it}	Other control variables, including Offensive Rating, Defensive Rating, and Pace, etc.

Hypothesis

We hypothesize that during the 2005–2024 NBA seasons, the marginal effect of 3-point attempt rate on team winning percentage first increased and later decreased, reflecting the rise and subsequent saturation of the modern small-ball era.

$$H_0 : \beta_2 = 0 \text{ and } \beta_3 = 0$$

$$H_1 : \text{not both zero}$$

(2)

We will use standard t-tests to see if β_2 and β_3 are significantly different from zero. If the null hypothesis is true, then there is no statistically significant time variation in the effect of three-point attempt rate (3PA%) on team winning percentage (WinRate).

Expected Results

We expect our data to provide sufficient evidence of the time-varying effect of three-point shot on team performance. Specifically, there is a non-linear effect that first goes up and then goes down over the period, indicating how NBA teams built and adapted to the rise and “saturation” of the small-ball era. We will calculate the marginal effect (first derivative) to test our hypothesis.

$$ME(t) = \frac{\partial WinRate_{it}}{\partial 3PA\%_{it}} = \beta_1 + \beta_2 \cdot YearIndex_t + \beta_3 \cdot YearIndex_t^2$$

3)

Potential Problems

Unobserved factors such as coaching strategies and player injuries may lead to omitted variable bias. Changes in league rules and style of play overtime could also create structural shifts that are not fully captured by the model. Finally, multicollinearity between interaction terms may increase standard errors and reduce the precision of coefficient estimates. We include team and season fixed effects, incorporate key control variables, and use robust standard errors to improve the reliability of our inference.

Results

The descriptive statistics show substantial variation in NBA team performance and three-point strategy from 2005 to 2024. Win percentage ranges widely, and 3PA/FGA varies from 0.23 to 0.52, indicating that teams differ meaningfully in how heavily they rely on three-pointers. In contrast, shooting efficiency varies less, suggesting that differences in shot selection drive much of the strategic variation.

Table 1. Summary Statistics (2005–2024)

Variable	Observations	Mean	Std. Dev.	Min	Max
Win%	600	0.500	0.150	0.106 (CHA, 2012)	0.890 (GSW, 2016)
3PA%	600	0.288	0.085	0.105 (LAC, 2005)	0.519 (HOU, 2019)
ORtg	600	108.56	4.39	95.20 (CHA, 2012)	123.20 (BOS, 2024)
DRtg	600	108.56	4.19	98.20 (BOS, 2012)	120.40 (UTA, 2024)
Pace	600	94.91	3.99	86.30 (MEM, 2006)	105.10 (MIL, 2020)

Note: Win% and 3PA% are expressed as proportions; Pace is possessions per 48 minutes; Def Rating is points allowed per 100 possessions (lower = better). Each observation is one team–season from 2005 to 2024.

The results in Table 2 show that the relationship between 3PA% and winning changes substantially once controls and fixed effects are introduced. For the quadratic model, Column (1) shows that on average, each percentage increase in 3PA% increases team win rate by 0.892%, and the negative (3PA%)² term suggests diminishing returns to 3PA%. After accounting for team and season fixed effects in Column (2), the coefficient on 3PA% falls to 0.025, indicating that teams do not win more simply because they take more threes.

Column (3) and (4) are the time interaction models. In the baseline season, a one percentage increase in team’s 3PA% results in a 0.042% decrease in winning percentage. However, the interaction with time indicates that in the early 2000s, a higher 3PA% was not meaningfully associated with winning (β_3, β_4 close to zero). Moving into the mid-2010s, as small-ball reached its competitive peak, increasing 3PA% became strongly linked to higher win percentages. By the early 2020s, however, the strategic advantage of simply taking more threes had begun to saturate, as league-wide adoption reduced the marginal benefit of 3-point volume alone.

Table 2. Regression Results of Quadratic and Interaction Models with Fixed Effects

	(1) <i>Quadratic</i> Win Rate	(2) <i>Quadratic + FE</i> Win Rate	(3) <i>Year-Interaction</i> Win Rate	(4) <i>Year-Interaction + FE</i> Win Rate
3PA%	0.892 (0.468)	0.025 (0.141)	-0.042 (0.056)	0.063 (0.119)
3PA% ²	-1.079 (0.783)	-0.157 (0.238)		
3PA% × YearIndex			0.004 (0.005)	-0.020 (0.029)
3PA% × YearIndex ²			-0.000 (0.000)	0.001 (0.001)
Offensive Rating		0.030*** (0.001)	0.030*** (0.001)	0.030*** (0.001)
Defensive Rating		-0.030*** (0.001)	-0.030*** (0.000)	-0.030*** (0.001)
Pace		-0.002* (0.001)	-0.001 (0.001)	-0.002 (0.001)
Constant	0.340*** (0.066)	0.596*** (0.121)	0.531*** (0.091)	0.586*** (0.120)
Team Fixed Effects	No	Yes	No	Yes
Season Fixed Effects	No	Yes	No	Yes
Observations	600	600	600	600

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	(1) <i>Quadratic</i>	(2) <i>FE +Interaction</i>
	Win Rate	Win Rate
3PA%	0.892	0.063
3PA% ²	-1.079	
3PA% × YearIndex		-0.020
3PA% × YearIndex ²		0.001
Offensive Rating		0.030***
Defensive Rating		-0.030***
Pace		-0.002
Constant	0.340***	0.586***
Team & Season FE	No	Yes
Observations	600	600